P2 Q4

$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\left(\frac{(x-\mu)^2}{2\sigma^2}\right)}$ $\quad X = \{1,3,5,7\}$ $\quad n = 4$

$L(\mu,\sigma^2) = \prod_{i=1}^{n} (2\pi\sigma)^{-\frac{1}{2}} e^{\left(\frac{(x_i-\mu)^2}{2\sigma^2}\right)}$

$\ell(\mu,\sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{(1-\mu)^2}{2\sigma^2} - \frac{(3-\mu)^2}{2\sigma^2} - \frac{(5-\mu)^2}{2\sigma^2} - \frac{(7-\mu)^2}{2\sigma^2}$

$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}(16 - 4\mu) = 0, \quad \boxed{\mu = 4}$ $\quad \frac{\partial^2 \ell}{\partial \mu^2} = \frac{-4}{\sigma^2}$ → concave, $\mu = 4$ is max

$\frac{\partial \ell}{\partial \sigma^2} = \frac{-4}{2\sigma^2} + \frac{20}{2\sigma^4} = 0 \rightarrow 80\sigma^{-4} = 40\sigma^{-2} \rightarrow \boxed{\sigma^2 = 5}$

P2 Q5

$P(\mu|x) \propto P(x|\mu)P(\mu)$ $\quad P(\mu|x) \propto e^{\left(-\frac{(5-\mu)^2}{8} - \frac{\mu^2}{2}\right)}$

$\frac{d}{d\mu}\ln P = \frac{(5-\mu)}{4} - \mu \rightarrow \mu = \frac{5-\mu}{4} \rightarrow 5\mu = 5 \rightarrow \boxed{\mu = 1}$

$\mu_{MAP} = 1$

P2 Q6

$\mathrm{Var}(X) = E\left((X - E(X))^2\right)$

$= E\left(X^2 - 2XE(X) + E(X)^2\right)$

$= E(X^2) - 2E[X]E(X) + E(X)^2$

$= E(X^2) - E(X)^2$

P3 Q1

$\mu = E[X]$

$\Sigma = \mathrm{cov}(X) = E[(X-\mu)(X-\mu)^T]$

$Y = u^T X$

$\mathrm{var}(Y) = E\left[(u^T X - E[u^T X])^2\right]$

$(u^T(X-\mu))^2 = u^T(X-\mu)(X-\mu)^T u$

$\mathrm{Var}(u^T X) = E[u^T(X-\mu)(X-\mu)^T u] =$

$= u^T E[(X-\mu)(X-\mu)^T] u =$

$= \boxed{u^T \Sigma u}$

P3 Q2

1. Two features have positive correlation of 2v.69 and their covariance is 1.2

2. Scatter plot is tilted up, meaning two features are positively correlated

P3 Q9

PC1 explains about 69% of data variance because features are strongly correlated so data is mostly along tilted axis

2. dimensionality reduction makes data smaller and easier to work with while only reducing variance by 11%.

P1 Q3

Say A, B are commutable → A+B=B+A

A is m×n and B is n×m →

AB is m×m and BA n×n →

$\text{Trace}(AB) = \sum_{i=1}^{m} A_{i,:} \cdot B_{:,i} = \sum_{i=1}^{m} a_{i1}b_{1i} + a_{i2}b_{2i} + \dots + a_{in}b_{ni} = \sum_{i=1}^{m}\left(\sum_{j=1}^{n}(a_{ij}b_{ji})\right)$

$\text{Trace}(BA) = \sum_{j=1}^{n} A_{j,:} \cdot B_{:,j} = \sum_{j=1}^{n} a_{j1}b_{1j} + a_{j2}b_{2j} + \dots + a_{mj}b_{jm} = \sum_{j=1}^{n}\left(\sum_{i=1}^{m}(a_{ij}b_{ji})\right)$

$n \cdot m = m \cdot n \longrightarrow \sum_{j=1}^{n}\sum_{i=1}^{m} a_{ij}b_{ji} = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}b_{ji} \rightarrow \text{Trace}(AB) = \text{trace}(BA)$

P2 Q1

a) $\sum_{k=0}^{\infty} c\frac{3^k}{k!} = 1 \rightarrow ce^3 = 1 \rightarrow c = e^{-3}$

$E(x) = e^{-3}\sum_{k=0}^{\infty}\frac{3^k}{k!}k = 3e^{-3}\sum_{k=1}^{\infty}\frac{3^{(k-1)}}{(k-1)!} = 3e^{-3}e^3 = 3$

$\text{Var}(x) = E(x^2) - 3^2$

~~$E(x^2) = \sum_{k=1}^{\infty}\left(\frac{3^k e^{-3}}{k!}\right)^2 k = e^{-3}\sum_{k=1}^{\infty} k\frac{3^k}{k!}\frac{3^k}{k!} = 3e^{-3}\sum_{k=1}^{\infty}\frac{3^{(k-1)}}{(k-1)!}\frac{3^k}{k!}$~~

$E(x^2) = e^{-3}\sum_{k=0}^{\infty} k^2\frac{3^k}{k!} \rightarrow e^{-3}\sum_{k=1}^{\infty} k\frac{3^k}{(k-1)!} \rightarrow j = k-1,\ 3e^{-3}\sum_{j=0}^{\infty}(j+1)\frac{3^j}{j!} \rightarrow 3e^{-3}\left(\sum_{j=0}^{\infty}\frac{3^j}{j!} + 3\sum_{j=1}^{\infty}\frac{3^j}{j!}\right) \rightarrow$

$3e^{-3}\cdot(e^3 + 3e^3) = 12$

$\text{Var}(x) = 12 - 9 = 3$

P2 Q3

$L(\theta) = \theta\cdot\theta\cdot(1-\theta)(1-\theta)\cdot\theta$

$L(\theta) = \theta^3(1-\theta)^2$

$\ell(\theta) = 3\ln\theta + 2\ln(1-\theta)$

$\frac{d\ell}{d\theta} = \frac{3}{\theta} - \frac{2}{1-\theta} \rightarrow 3 - 3\theta = 2\theta \rightarrow 3 = 5\theta$

MLE $= \frac{3}{5}$

$\frac{d^2\ell}{d\theta^2} = -\frac{3}{\theta^2} - \frac{2}{(1-\theta)^2}$ → negative for all $\theta \in (0,1)$

P1 Q2

$P(A) = 0.5 \quad P(\theta) = 0.3 \quad P(A|\theta) = 0.6$

$P(\theta|A) = \frac{0.6\cdot 0.3}{0.5} = 0.36$

$P(A \text{ and } \theta) = 0.6\cdot 0.3 = 0.18$

$P(A \text{ or } \theta) = 0.5 + 0.3 - 0.18 = 0.62$

$P(\theta|D) \propto \theta^4(1-\theta)^2$

$\frac{d}{d\theta}\ln(P(\theta|D)) = \frac{4}{\theta} - \frac{2}{1-\theta}$

$4 = 6\theta \quad \theta = \frac{2}{3}$

MAP $= \frac{2}{3}$

HW 1

P1Q2

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \lambda\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\begin{bmatrix} 2v_1 - v_2 \\ -v_1 + 2v_2 \end{bmatrix} = \begin{bmatrix} \lambda v_1 \\ \lambda v_2 \end{bmatrix}$$

$$2v_1 - v_2 = \lambda v_1$$
$$-v_1 + 2v_2 = \lambda v_2$$
$$v_1 + v_2 = \lambda v_1 + \lambda v_2$$

$$\det\left(\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}\right) \Rightarrow 3 - 4\lambda + \lambda^2 = 0 \Rightarrow (\lambda - 1)(\lambda - 3) = 0$$

$\lambda = 1, 3$

$\lambda = 1$

$(2-\lambda)v_1 - v_2 = 0$

$-v_1 + (2-\lambda)v_2 = 0$

$v_1 - v_2 = 0$

$-v_1 + v_2 = 0$

$v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}, v_1 = v_2, v \in \mathbb{R}$

$\lambda = 3$

$-v_1 - v_2 = 0$

$-v_1 - v_2 = 0$

$v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}, v_1 \in \mathbb{R}, v_1 = -v_2, v_1 \neq 0$

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ -3 \end{pmatrix} = 3 \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

P1Q1

$$A \cdot B = \begin{pmatrix} 3 & 4 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 3 & -2 \\ 2 & 3 \end{pmatrix}$$

$$B \cdot A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 3 & 4 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 7 & 6 \\ -2 & -1 \end{pmatrix}$$

$$A + B = \begin{pmatrix} 4 & 6 \\ 2 & 0 \end{pmatrix}, \quad (A+B)^T = \begin{pmatrix} 4 & 2 \\ 6 & 0 \end{pmatrix}$$

$$B^T = \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix} \quad AB^T = \begin{pmatrix} 3 & 4 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 11 & -4 \\ 4 & -1 \end{pmatrix}$$

- $\text{Trace}(A) = 3 + 1 = 4$
  $\text{Trace}(AB) = 3 + 3 = 6$

P1Q

$$\det(C) = 24 - 14 = 10$$

$$C^{-1} = \frac{1}{10}\begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}\begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} =$$